ACML100

Lecture 6

Determination of rate law

- Isolation method: Add large excess
- Methods of initial rate: measurement of rate of reaction for several different initial concentration of reactants.

Isolation Method

- Suppose a reaction involves two reactants, A and B.
- $\vartheta = k[A]^{\alpha}[B]^{\beta}$
- To determine the order of reaction with respect to A, we will take B in large excess. $[B] \approx [B]_0$

$$\vartheta = k[A]^{\alpha}[B]_0^{\beta}$$

$$\vartheta = k'[A]^{\alpha}$$

Methods of Initial Rates

- Suppose a reaction involves two reactants A and B.
- $\vartheta = k[A]^{\alpha}[B]^{\beta}$

- The rate of reactions is determined at initial stage of the reaction where $[A] \sim [A]_0$ and $[B] \sim [B]_0$.
- $\vartheta = k{[A]_0}^{\alpha}{[B]_0}^{\beta}$

Methods of Initial Rates

- Change the initial concentration of A keeping the initial concentration of B constant

$$\vartheta' = k[A]_0'^{\,\alpha}[B]_0^{\,\beta}$$

$$\frac{\vartheta'}{\vartheta} = \left(\frac{[A]_0'}{[A]_0}\right)^{\alpha}$$

Methods of Initial rates

- If the reaction is of first order with respect to A
- $\frac{\vartheta'}{\vartheta} = \left(\frac{[A]_0'}{[A]_0}\right)$
- If the reaction is of second order with respect to A
- $\frac{\vartheta'}{\vartheta} = \left(\frac{[A]_0'}{[A]_0}\right)^2$

Integrated Rate law in terms of Physical Parameter

- Total pressure of a gaseous reaction
- Total volume of a reaction involving liquids
- Optical rotation of the reacting sample involving optically active compound
- Conductivity of a reacting sample involving ions.
- Absorbance of reacting sample involving compounds absorbing in UV-Vis range.

$$NH_3(\mathrm{g}) \quad \rightarrow \quad \frac{1}{2}N_2(g)+\frac{3}{2}H_2(g)$$

@ time t = 0	a	0	0	$\mathrm{P_0} = \mathrm{k_1 a}$
@ time t = t	(a - x)	$x/2$	$3\,x/2$	$\mathrm{P_t} = \mathrm{k_1}(a+x)$
@ time $t = \infty$	0	$a/2$	$3a/2$	$\mathrm{P_\infty} = \mathrm{k_1}\,2a$

$$P_t - P_\infty = -k_1(a-x)$$

$$P_0 - P_\infty = -k_1 a$$

$$\frac{P_t - P_\infty}{P_0 - P_\infty} = \frac{(a-x)}{a}$$

$$A \rightarrow P$$

@ time $t = 0$	a	0	$Y_0 = k_1 a$
@ time $t = t$	$(a - x)$	x	$Y_t = k_1(a - x) + k_2 x$
@ time $t = \infty$	0	a	$Y_\infty = k_2 a$

$$Y_t - Y_\infty = k_1(a - x) - k_2(a - x) = (k_1 - k_2)(a - x)$$

$$Y_0 - Y_\infty = (k_1 - k_2)a$$

$$\frac{Y_t - Y_\infty}{Y_0 - Y_\infty} = \frac{(a - x)}{a}$$

$$A \quad + \quad 3B \quad \rightarrow \quad 2C \quad + \quad D$$

@ time $t = 0$	a	b	0	0	$Y_0 = k_1 a + k_2 b$
@ time $t = t$	$(a - x)$	$(b - 3x)$	$2x$	x	$Y_t = k_1(a-x) + k_2(b-3x) + k_3(2x) + k_4(x)$
@ time $t = \infty$	0	$(b - 3a)$	$2a$	a	$Y_\infty = k_2(b-3a) + k_3(2a) + k_4 a$

$$Y_t - Y_\infty = -(a-x)(2k_3 + k_4 - k_1 - 3k_2)$$
$$Y_0 - Y_\infty = -a(2k_3 + k_4 - k_1 - 3k_2)$$

$$\frac{Y_t - Y_\infty}{Y_0 - Y_\infty} = \frac{(a-x)}{a}$$

For a first order reaction

$$ln\frac{[A]_0}{[A]} = ln\frac{a}{(a-x)} = kt$$

$$ln\frac{[A]_0}{[A]} = ln\frac{Y_0 - Y_\infty}{Y_t - Y_\infty} = kt$$

For a second order reaction

$$\frac{1}{[A]} - \frac{1}{[A]_0} = kt$$

$$\frac{[A]_0}{[A]} - 1 = kt[A]_0$$

$$\frac{[A]_0}{[A]} = kt[A]_0 + 1$$

$$\frac{\boldsymbol{Y_0 - Y_\infty}}{\boldsymbol{Y_t - Y_\infty}} = kt[A]_0 + 1$$

Guggenheim Method

$$ln\left(\frac{P_t - P_\infty}{P_0 - P_\infty}\right) = -kt$$

$$\left(\frac{P_t - P_\infty}{P_0 - P_\infty}\right) = exp(-kt)$$

$$(P_t - P_\infty) =$$
$$(P_0 - P_\infty)exp(-kt)$$

$$P_t = (P_0 - P_\infty)exp(-kt) + P_\infty$$

$$P_{t+\tau} = (P_0 - P_\infty)exp[-k(t + \tau)] + P_\infty$$

$$P_t - P_{t+\tau} =$$
$$(P_0 - P_\infty)\big(1 - exp(-k\tau)\big)exp(-kt)$$

$$ln(P_t - P_{t+\tau}) =$$
$$ln(P_0 - P_\infty) + ln\big(1 - exp(k\tau)\big) - kt$$

Kezdy Swinbourne Method

$$(P_t - P_\infty) = (P_0 - P_\infty)exp(-kt)$$

$$P_{t+\tau} - P_\infty = (P_0 - P_\infty)exp[-k(t+\tau)]$$

$$\frac{(P_t - P_\infty)}{(P_{t+\tau} - P_\infty)} = exp(k\tau)$$

$$(P_t - P_\infty) = (P_{t+\tau} - P_\infty)\ exp(k\tau)$$

$$P_t = \left(1 - exp(k\tau)\right)P_\infty + P_{t+\tau}exp(k\tau)$$